

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the)
Interstate Transmission System)
)

Docket No. RM26-4-000

**INITIAL ANOPR COMMENTS OF THE WORKING FOR ADVANCED
TRANSMISSION TECHNOLOGIES AND ADVANCING MODERN POWERLINES
COALITIONS**

I. Introduction

The Working for Advanced Transmission Technologies (WATT) Coalition and the Advancing Modern Powerlines (AMP) Coalition thank the Federal Energy Regulatory Commission (“FERC” or “the Commission”) for the opportunity to provide comments in response to the Secretary of Energy’s (“Secretary”) proposed advanced notice of proposed rulemaking (“ANOPR”) regarding the interconnection of large loads to the interstate transmission system pursuant to the Secretary’s authority under section 403 of the Department of Energy Organization Act¹ published by the Secretary on October 27, 2025.

We agree with the Secretary that the United States is facing rapid growth in energy demand from large loads, in particular, data centers and advanced manufacturing, and that in order to for the United States to prosper, domestic industries and citizens alike must have access to affordable, reliable, and secure electricity.² We commend the Secretary and the Administration for its focus on the transmission system and benefits of Advanced Transmission Technologies (“ATTs”).³

In particular, the WATT and AMP Coalitions are submitting comments to highlight key points for consideration by the Commission.

1. The Commission found in Order No. 2023 that ATTs can provide benefits to generator interconnection customers network upgrades in place of or in addition to traditional network upgrades. The Commission required the evaluation of ATTs in the

¹ 42 U.S.C. § 7173.

² Advanced Notice of Proposed Rulemaking, “Interconnection of Large Loads to the Interstate Transmission System,” Docket No. RM26-4-000 (Oct. 23, 2025) (“ANOPR”).

³ In outlining his priorities as Secretary of Energy, Chris Wright stated that the nation’s transmission system is “critical to the reliable and secure delivery of electricity.” *See* U.S. Secretary of Energy, Secretarial Order: Unleashing the Golden Era of American Energy Dominance” (Feb. 5, 2025), <https://www.energy.gov/articles/secretary-wright-acts-unleash-golden-era-american-energy-dominance>; The White House’s AI Action Plan includes a section on grid optimization, specifically highlighting as a priority “upgrades to power lines that can increase the amount of electricity transmitted along existing routes.” *See* Executive Office of the President of the United States, Winning the Race: America’s AI Action Plan, at 15 (July 2025), <https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf>.

interconnection study process because they are cheaper and faster to deploy than traditional upgrades, saving money and time.

2. The benefits the Commission found for generator interconnection apply for large load interconnection as well and are key to achieving speed to power: ATTs are **the fastest way to increase transmission capacity** and provide additional benefits to large loads and consumers through lower upgrade and system costs, as well as improved reliability.
3. The WATT and AMP Coalitions recommend the Commission require ATTs, including dynamic line ratings, be evaluated alongside traditional upgrades in the interconnection of large loads, as was required in Order No. 2023, given the benefits to consumers and interconnection customers identified by the Commission.

The WATT and AMP Coalitions do not provide comment on any of the other specific issues raised in the ANOPR, other than to respectfully request that the evaluation of ATT deployment to facilitate large load interconnections should be considered as a principle that informs any rulemaking procedures proposed by the Commission.

II. About the WATT Coalition and AMP Coalition

Grid Enhancing Technologies (“GETs”) and High Performance Conductors (“HPCs”), together known as Advanced Transmission Technologies (“ATTs”),⁴ represent the fastest, lowest-cost path to unlock grid capacity and enable the interconnection of new generation and AI datacenters, and other large loads.

a. About the WATT Coalition

The WATT Coalition advocates for policy that supports the wide deployment of GETs to accelerate lower energy costs and enable economic growth. GETs are hardware and/or software that increases the capacity, efficiency, and maintain the reliability of the electric grid. Learn more at watt-transmission.org. WATT’s membership includes Dynamic Line Ratings, Advanced Power Flow Control and Transmission Topology Optimization vendors, as well as independent power producers, investors, and utilities.

i. About Grid Enhancing Technologies

GETs are hardware and software that increase the capacity, efficiency and reliability of the transmission system. These technologies are installed on existing transmission infrastructure to give operators more situational awareness and control over the grid.

⁴ The WATT and AMP Coalitions use “Advanced Transmission Technologies” to refer collectively to the GETs and HPCs included in the section which the Commission describes as “alternative transmission technologies.” For clarity, the term “ATT” is used to encompass both “Advanced Transmission Technologies” and “alternative transmission technologies.”

- **Advanced Power Flow Controller (“APFC”) technologies** actively balance the flow on transmission lines. The hardware can intelligently raise or lower the impedance, or the opposition to current, in real time to push power from overloaded circuits to lines that have the capacity to carry it.
- **Dynamic Line Rating (“DLR”)** monitors ambient conditions, including wind speed, which heat or cool transmission lines to calculate the true capacity of transmission lines based on their thermal limits. DLR often allows significantly more power flow than a static rating over the course of the year but also detects when flows should be reduced to continue safe and reliable operation in extreme heat or other conditions.
- **Transmission topology optimization (“TTO”)** software models the grid's network and power flow conditions to identify ways to reroute power flow around congested or overloaded transmission elements. These "reconfigurations" are implemented by switching on or off existing high-voltage circuit breakers. By more evenly distributing flow over the network, topology control increases the transfer capacity of the grid, like Google maps redirecting cars in traffic.

b. About the AMP Coalition

The AMP Coalition is an ad hoc coalition of HPC technology vendors. AMP’s goal is to further the use of HPCs as a tool for modernizing and increasing grid capacity, as well as improving the overall resilience, reliability, and energy efficiency of the grid. The coalition includes both carbon core conductors and superconductors. Learn more at ampcoalition.org.

i. About High Performance Conductors

HPCs are modern conductor technologies which have greater performance characteristics when compared to traditional conductors, including increased capacity, higher efficiency, and less thermal sag.

- **Carbon Core Conductors**, also known as Advanced Conductors, are overhead, bare conductors that use a trapezoid shaped wire of annealed aluminum to carry electrical current and use a carbon core for support. Used commercially for 20 years, the conductors are deployed in over 60 countries across five continents.⁵ Carbon core conductors have three key advantages over traditional conductors: 1) stronger and lighter weight cores, which allows for more aluminum to be added to the conductor, doubling the capacity; 2) 20% or greater efficiency; and 3) half as much thermal sag.⁶

⁵ ACORE, Grid Strategies, *Unlocking the Grid: A Playbook on High Performance Conductors for State and Regional Regulators and Policymakers* (Oct. 2024), at 2, <https://acore.org/wp-content/uploads/2024/10/Unlocking-the-Grid-A-Playbook-on-High-Performance-Conductors-for-State-and-Regional-Regulators-and-Policymakers.pdf> (“HPCs Playbook”).

⁶ *Id.*

- **Superconductors** use a class of metallic compounds that exhibit negligible resistance when cooled using liquid nitrogen, enabling very low losses and very high power-flow capacities. Developed in the 1980s, countries in Europe and Asia, as well as the U.S., have installed superconductors over the past 20 years.⁷ Superconductors have three key advantages over traditional conductors: 1) increase capacity 5-10 times at lower voltages, reducing substation build and cost; 2) no thermal sag and lines do not vary with ambient weather conditions because there is no exposure to elements, or can be undergrounded in small ROWs; and 3) at least 50% more energy efficient.⁸

III. Transmission interconnection is critical to the development of large loads

The rapid growth in electricity use to train and operate increasingly powerful AI models is putting significant pressure on the U.S. power system. A 2024 Department of Energy analysis projects that data centers could account for up to 12% of total U.S. electricity consumption by 2028, up from about 4.4% in 2023.⁹

Data center buildout in PJM is expected to host as much as 70% of future U.S. development.¹⁰ The concentration of demand, especially in northern Virginia’s “data center alley,” where a substantial share of global internet traffic flows, is creating intense pressure for new greenfield transmission to support the interconnection of data centers.¹¹ Building those lines can be challenging. New lines take many years to build, are capital intensive, must cross numerous local jurisdictions, and can involve disputes over land use and eminent domain. New greenfield transmission projects can be contentious – a fundamental reality that can be managed but not eliminated. Consequently, these dynamics have led to utilities in some cases quoting wait times as long as seven years for new large loads to interconnect.¹²

Additionally, behind-the-meter (“BTM”) generation options can be beneficial but do not solve all problems for data center operators, who have turned to a wide array of agreements with generation developers to accelerate the powering of these facilities. But, for now, most data

⁷ *Id.*

⁸ ENTSO-E, Technopedia: High Temperature Superconducting (HTS) Cables (Mar. 2025), <https://www.entsoe.eu/technopedia/techsheets/high-temperature-superconducting-cables/> (hereinafter, “Technopedia”).

⁹ Lawrence Berkeley National Laboratory, “2024 United States Data Center Energy Usage Report,” December 2024, <https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf> at 5.

¹⁰ Boston Consulting Group, “Breaking Barriers to Data Center Growth,” January 20, 2025, <https://www.bcg.com/publications/2025/breaking-barriers-data-center-growth>.

¹¹ Prince William Times, “Opposition mounts to ‘spiderweb’ of new transmission lines needed to power data center alley,” October 25, 2023, https://www.princewilliamtimes.com/news/opposition-mounts-to-spiderweb-of-new-transmission-lines-needed-to-power-data-center-alley/article_b7da863c-dfac-516d-9dfe-9adc34b8c8bd.html.

¹² Bloomberg, “Data Centers Face Seven-Year Wait for Dominion Power Hookups,” August 29, 2024, <https://www.bloomberg.com/news/articles/2024-08-29/data-centers-face-seven-year-wait-for-power-hookups-in-virginia?embedded-checkout=true>

centers still seek a grid interconnection, even if some generation is co-located. Eliminating grid interconnection entirely creates risks which many large load interconnection customers are not willing to accept. AI data centers seek to operate at high levels of reliability, often at least 99.999 percent (‘five nines’) uptime.¹³ That standard allows for less than seven minutes of outage in a year. Any off-grid generation solution would require maintenance outages exceeding that limit and offer less redundancy compared to the grid. For the vast majority of operators, grid interconnection with generator backup is preferable to fully off-grid data centers.

IV. ATTs are the fastest option to unlock grid capacity and interconnect new data centers and other large loads

“Speed-to-power” has emerged as the key priority for data center developers, outpacing other considerations such as access to land, water, chips, or fiber availability, and the price of electricity.¹⁴ However, development of new transmission lines requires, on average, six to ten years.¹⁵ While new greenfield transmission projects will be needed in some cases to support the interconnection of large loads, ATTs deployment timelines are more aligned with the interconnection needs of new large loads, which may only take 1-2 years to construct. ATTs can rapidly increase the grid’s capacity in 18-36 months using existing infrastructure and rights-of-way (“ROW”) providing fast solutions to unlock new grid capacity while also being a bridge to bring new large loads on more quickly while new network transmission upgrades are built.

Within 1-3 months, DLR and TTO can be deployed. Belgium’s system operator Elia has operated with DLR since 2012, finding 10-16% percent or more capacity gains 90% of the time.¹⁶ Deploying TTO for a utility in South America found 7% additional transfer capacity. A 7% increase in transfer capacity in the US could mean around 100 gigawatts (“GW”) of additional capacity.

Within 1-3 years, APFC can be deployed and can add hundreds of megawatts (“MW”) of additional capacity. In the United Kingdom, APFC unlocked 2 GW of additional capacity across

¹³ SemiAnalysis, “Datacenter Anatomy Part 1: Electrical Systems,” October 24, 2024, <https://semianalysis.com/2024/10/14/datacenter-anatomy-part-1-electrical/>.

¹⁴ McGeady, Cy et al., “The Electricity Supply Bottleneck on US AI Dominance,” March 2025, https://csis-website-prod.s3.amazonaws.com/s3fs-public/2025-03/250303_McGeady_Electricity_Bottleneck.pdf?VersionId=LsCkrctQBekLSYFPxus3XWdmTCnmIr9Z

¹⁵ See MISO, [LRTP Tranche 1 Appendix A](#) (2022) and [LRTP Tranche 2.1 Appendix A](#); See also Lisa Kelso, Testimony before the Public Utilities Commission of Ohio, “Direct Testimony of Lisa Kelso on Behalf of Ohio Power Company,” Case No. 24-508-EL-ATA, filed May 13, 2024, <https://dis.puc.state.oh.us/ViewImage.aspx?CMID=A1001001A24E13B43501B00952>.

¹⁶ Ampacimon, Inc., *Elia Large-scale DLR Deployment* (Sept. 2020), <https://ferc.gov/sites/default/files/2020-09/Alexander-ELIA.pdf>.

three substations, and other individual deployments in the United States have found 100-500 MWs of capacity.¹⁷

HPCs can be deployed on a similar 1-3 year timeframe through reconductoring and can double capacity on an existing ROW.¹⁸ ENEL in Colombia doubled line capacity when reconductoring 70 kilometers with HPCs, alleviating congestion to support the growth of and increased energy demands from industry.¹⁹ SSEN in Scotland reconductored 44 kilometers of circuit miles with HPCs, doubling line capacity to meet demand from regional heavy industry, such as an aluminum smelter.²⁰

V. The Commission recognized the benefits of ATTs to the interconnection process in Order No. 2023

The Commission has recognized that omitting ATTs from the generation interconnection study processes can lead to unjust and unreasonable rates. In Order No. 2023 *Improvements to the Generator Interconnection Procedures and Agreements* the Commission found that

“Because the benefits of the enumerated alternative transmission technologies identified above are present across all interconnection processes, regardless of the size of the interconnection request, we find that the failure to evaluate the enumerated alternative transmission technologies results in both the pro forma LGIP and pro forma SGIP being unjust, unreasonable, and unduly discriminatory or preferential.”²¹

The Commission also acknowledged the value of ATTs in lowering upgrade costs and shortening timelines. Order No. 2023 found that ATTs “can be deployed more quickly to be used as network upgrades in place of, and at a lower cost than, traditional network upgrades,” and highlighted the numerous benefits commenters stated ATTs can provide including, “resolving thermal overloads and avoiding voltage collapse, allowing for better use of the existing transmission system, improving reliability, and reducing interconnection request withdrawals, restudies, and overall interconnection delays.”²²

¹⁷ Smart Wires Inc., *Case Study Booklet*, accessed November 14, 2025, <https://watt-transmission.org/wp-content/uploads/2025/11/Smart-Wires-case-study-booklet.pdf>.

¹⁸ AMP Coalition & ACORE, *Unlocking the Grid: A Playbook on High Performance Conductors* (October 2024), <https://acore.org/wp-content/uploads/2024/10/Unlocking-the-Grid-A-Playbook-on-High-Performance-Conductors-for-State-and-Regional-Regulators-and-Policymakers.pdf>.

¹⁹ CTC Global, *ENEL Reconductors to Ensure a Reliable Energy Future*, accessed November 14, 2025, <https://ctcglobal.com/enel-reconductoring-case-study/>.

²⁰ CTC Global, *SSEN Reconductors for Added Capacity in the Scottish Highlands*, accessed November 14, 2025, <https://ctcglobal.com/ssen-reconductoring-case-study/>.

²¹ *Improvements to Generator Interconnection Procedures and Agreements*, 184 FERC ¶ 61,054 (2023) at P 54 (“Order No. 2023”).

²² *Id.*

FERC now requires ATTs to be evaluated for every generation interconnection customer in every cluster study. In Order No. 2023, the Commission required transmission providers:

“Evaluate the list of alternative transmission technologies enumerated in this final rule during the cluster study, including any restudies, of the generator interconnection process in all instances (i.e., for all interconnection customers in a cluster), without the need for a request from an interconnection customer. We require transmission providers to evaluate each alternative transmission technology listed in pro forma LGIP section 7.3 and to determine, in the transmission provider’s sole discretion, whether it should be used, consistent with good utility practice, applicable reliability standards, and other applicable regulatory requirements. Finally, we require transmission providers to include, in the pro forma LGIP cluster study report, an explanation of the results of the evaluation of the enumerated alternative transmission technologies for feasibility, cost, and time savings as an alternative to a traditional network upgrade.”²³

In Order No. 2023 the Commission also identified specific benefits ATTs contribute to the generation interconnection process:

- “Regarding advanced power flow controls, we find that these devices allow power to be pushed and pulled to alternate lines with spare capacity leading to maximum utilization of transmission capacity and mitigation of overloads. Advanced power flow control devices can be scaled back as needed, providing an advantage over new lines or reconductors.”²⁴
- “Transmission switching can be used to route energy around areas with high congestion and improve the overall transfer capability of the system, potentially resulting in lower network upgrade costs.”²⁵
- “We note the comments arguing that advanced conductors may be beneficial as network upgrades...deploying advanced conductors can significantly increase transmission capacity and allow for the interconnection of new generating facilities without the construction of new network upgrades...By increasing transmission line ratings, there will be more ‘headroom’ on the system to address normal and contingency conditions identified in interconnection studies, and likely a reduced need for network upgrades. Given these potential benefits to interconnection customers, we require transmission providers to evaluate advanced conductors and tower lifting in the interconnection process.”²⁶

The WATT and AMP Coalitions respectfully request that the list of ATTs required to be studied in Order No. 2023 should also be required alongside the other principles proposed by the

²³ *Id.* at P 1578.

²⁴ *Id.* at P 1594

²⁵ *Id.* at P 1583

²⁶ *Id.* at P 1597.

Secretary for any process for the interconnection of large loads to the interstate transmission system.

VI. Inclusion of DLR in the context of large load interconnection

The WATT Coalition respectfully recommends that DLR should be evaluated in the context of any new large load interconnection procedures alongside other ATTs.

DLRs were initially included on the ATT list in the Notice of Proposed Rulemaking (“NOPR”) for Order No. 2023.²⁷ However, DLR was removed from the list of ATTs required to be evaluated in Order No. 2023, ostensibly because it was not among the technologies showing the “most potential to be useful,”²⁸ despite a well-developed record showing benefits in the generator interconnection process. DLRs have been demonstrated to be effective interconnection-related network upgrades, such as unlocking 600 MW of wind generation in the UK,²⁹ reduce the need for more expensive and longer-lead time traditional network upgrades and facilitate more rapid interconnection. DLRs accurately forecast and confirm the true, real-time capacity of power lines by including various inputs such as ambient air temperature or transmission line tension.

In the interconnection study process, upgrades may be identified based on very small overloads on lines with extremely conservative thermal ratings. One generation developer was assessed for a \$400 million upgrade based on a 1% projected line overload. Depending on the utility’s rating methodology, DLR has potential to increase the capacity of that line by more than 1% in all conceivable weather scenarios. While DLR will not be appropriate for every load interconnection, it will occasionally save many millions of dollars and years of delay and thus should be studied. If the interconnection agreement includes load flexibility, DLR will be even more valuable in reducing the number of hours requiring load curtailment.

DLR regularly provides potential as a cost-effective solution in interconnection processes. The WATT Coalition in its reply comments to the Order No. 2023 NOPR cited a substantial record supporting DLR in Appendix A and in Appendix B provide additional specific examples of the use of DLR in the generator interconnection context.³⁰ The WATT Coalition’s request for rehearing and clarification of Order No. 2023 provided multiple further examples from around

²⁷ Notice of Proposed Rulemaking, *Improvements to Generator Interconnection Procedures and Agreements*, 179 FERC ¶ 61,194, at P 294-295 (2022) (“NOPR”).

²⁸ Order No. 2023 at P 1579.

²⁹ National Grid Partners, “Upgrading the grid,” accessed November 21, 2025, <https://www.ngpartners.com/stories/how-national-grid-partners-is-working-to-achieve-net-zero-emissions-by-20502>.

³⁰ Working for Advanced Transmission Technologies Reply Comments on Improvements to Generator Interconnection Procs. & Agreements, 87 FR 39934 (July 5, 2022), 179 FERC ¶ 61,194, at 7-17 (2022) https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20221214-5150&optimized=false&sid=48cb2214-3438-40f2-bc0b-f88dcfabac3b (“WATT Reply Comments”).

the world where generators around the world have benefited from the inclusion of DLR in interconnection upgrades.³¹

VII. Conclusion

The WATT and AMP Coalitions commend the Secretary for its work on the ANOPR and the Commission for providing opportunity to comment. We reiterate the need for the swift deployment of ATTs to help facilitate the interconnection of large loads.

We also emphasize that our comments do not contain recommendations on additional issues raised by the Secretary's proposed ANOPR, beyond our recommendation that any rulemaking procedures proposed by the Commission should include the evaluation of ATT deployment to facilitate large load interconnections.

Additionally, the Commission could issue a separate rulemaking on the use of all ATTs to address issues such as congestion or grid efficiency and utilization in near term planning and operations.

We welcome continued engagement with the Secretary and Commission to help identify and accelerate practical solutions that align with national goals of large load interconnection.

Respectfully submitted,

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³¹ Working for Advanced Transmission Technologies Request for Rehearing and Clarification on Improvements to Generator Interconnection Procs. & Agreements, 184 FERC ¶ 61,054 (2023), https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20230828-5257&optimized=false&sid=48cb2214-3438-40f2-bc0b-f88dcfabac3b.